

EE5203 L06 Pre-Lab Quiz

From polling to interrupts - Basri Erdogan

Preparation time: 5-8 minutes **Attempts:** Two; highest score recorded **Bring to the lab:** Your answer to Question 5

1. Pressing B1 produces which edge on PC13, and why?
 - a. Rising — the button connects PC13 to 3.3 V.
 - b. Falling — the external pull-up holds PC13 high until the button grounds it.
 - c. Falling — the internal pull-down releases the pin.
 - d. Rising — interrupts always use rising edges.
2. Which function does the **processor** enter automatically, and which one does **HAL** call afterwards?
 - a. Processor: `HAL_GPIO_EXTI_Callback` → HAL: `EXTI15_10_IRQHandler`
 - b. Processor: `EXTI15_10_IRQHandler` → HAL: `HAL_GPIO_EXTI_Callback`
 - c. Processor: `main()` → HAL: `EXTI15_10_IRQHandler`
 - d. Both are called from the `while (1)` loop.
3. The NVIC checkbox for `EXTI line[15:10] interrupts` is left unchecked. What happens on a press?
 - a. Nothing at all — EXTI never notices the edge.
 - b. The callback runs but with the wrong pin argument.
 - c. `EXTI->PR` bit 13 is set, but the request is never delivered to the CPU — no handler, no callback.
 - d. The program crashes into the fault handler.
4. Why must `button_event` be declared `volatile`?
 - a. It makes the variable atomic.
 - b. Interrupt code changes it between main-loop instructions, so the compiler must re-read it instead of using a stale copy.
 - c. It moves the variable into a hardware register.
 - d. HAL requires all globals to be volatile.
5. Complete this entry ticket before class:

Prediction	Your answer
EXTI->PR bit 13 at a breakpoint inside <code>EXTI15_10_IRQHandler</code> The register field that routes PC13 (not PA13/PB13) to <code>EXTI13</code> Edges at t = 5000, 5003, 5012, 5240 ms — which does the 80 ms guard accept? One thing that must never appear inside the callback	